

SAFIUS SAKIB SHUDDHO

s.shuddho@gmail.com linkedin.com/in/sshuddho github.com/sshuddho

RESEARCH INTERESTS: Particle Physics, Quantum Field Theory, Beyond-the-Standard-Model Physics, Dark Matter Phenomenology

EDUCATION

Master of Science (MSc) in Theoretical Physics (Ongoing) **Mar 2026 – Present**
University of Dhaka, Dhaka, Bangladesh

Courseworks: Quantum Mechanics, Statistical Mechanics, Quantum Field Theory, Introduction to Particle Physics, Introduction to Condensed Matter Physics, General Relativity and Cosmology

Bachelor of Science (BSc) in Nuclear Engineering **Apr 2021 – Jul 2025**
Military Institute of Science and Technology (MIST), Dhaka, Bangladesh
CGPA: 3.49/4.00

COMPUTATIONAL PROJECTS

1. DecayMC:

A Monte Carlo Framework for Relativistic Particle Decays and Scattering. **Report**

- Implemented Lorentz-covariant four-vector kinematics, two-body and three-body decays, Michel-weighted muon decay, and Rutherford scattering.

RESEARCH EXPERIENCE

UNDERGRADUATE THESIS

Department of Nuclear Science & Engineering, Military Institute of Science and Technology, Dhaka.

Title: NEUTRONICS CHARACTERISTICS ANALYSIS OF ALFRED WITH LFR REFLECTOR COMPARISON AND EVALUATION OF A CONCEPTUAL MPRR USING OPENMC [View Thesis]

Supervisor: Prof. Dr. Abdus Sattar Mollah ([Email](#))

PUBLICATIONS

- Shuddho, S. S., Sunny, A. A., & Mollah, A. S., "Neutronic Performance of Reflector Materials in Lead-cooled Fast Reactor," **Nuclear Science and Engineering**, Apr 2026. DOI: 10.1080/00295639.2026.2652786.

CONFERENCES

- Shuddho, S. S., Sunny A. A., Mollah A. S., "Neutronic Evaluation of Reflector Materials for the ALFRED Core.", *Second Joint ICTP-IAEA Workshop on Open-Source Nuclear Codes for Reactor Analysis*, Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy, Sep 2025. **Role:** Presented a Poster [Poster PDF].
- Dipto R. R., Shuddho, S. S., Sunny A. A., Mollah A. S., "Analysis of Neutronics Parameters of Different Annular Fuel using Monte Carlo Code OpenMC Utilizing JEFF-3.3 and ENDF/B-VIII.0 Nuclear Data Libraries", *Energy Conference 2023: National and Global Issues (ENCON23)*., IESD, BUET, Dhaka, Dec 2023. DOI: 10.2139/ssrn.4997514. **Role:** Attended as an Author.

SELECTED TRAINING AND WORKSHOPS

1. **Undergraduate Winter School on Particle Physics** [Certificate] **Nov 2025**
 - Completed an intensive lecture series covering fundamentals of particle physics, neutrino mass and mixing, CP violation, sterile neutrinos, and neutrinoless double-beta decay.
 - Gained foundations on Feynman diagrams, particle accelerator principles, detector technologies, and the design and interpretation of modern high-energy physics experiments.

2. **Second Joint ICTP–IAEA Workshop on Open-Source Nuclear Codes for Reactor Analysis**, [Certificate]
Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy Sep 2025
 - Presented a poster: “*Neutronic Evaluation of Reflector Materials for the ALFRED Core.*”
 - Participated in hands-on sessions with OpenMC, OpenFOAM, MOOSE, and Cardinal.
3. **Fundamentals of Reactor Physics with Python**,
International Atomic Energy Agency (IAEA) Dec 2024 - Oct 2025
 - Covered reactor physics fundamentals such as neutron slowing down, multi-group neutron transport, point reactor kinetics, and fuel depletion.
Textbook: *Nuclear Reactor Analysis* (Duderstadt & Hamilton, 1976).
 - Completed 11 guided datalabs and 8 independent assignments in Python.
 - Scored **98.08%**, top 8%.
4. **MTV Nuclear Engineering Summer School**, [Certificate]
Consortium for Monitoring, Technology, and Verification Jun - Aug 2023
 - Studied nuclear structure, radiation physics, detection & imaging, and fission science.
 - Attended lectures on Monte Carlo methods, nuclear data evaluation, and experiment design.
 - Explored advanced applications such as non-destructive assay, radioxenon monitoring, radiation protection, radiotherapy planning, and nuclear science outreach.
5. **CS50: Introduction to Computer Science**, Harvard University [Certificate] Jan - Dec 2023
 - Gained proficiency in C, Python, SQL, JavaScript, and web technologies while applying concepts such as data structures, resource management, and security.
 - Completed 10 rigorous problem sets, 9 labs, and delivered a final programming project.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, MATLAB, SQL
- **Computational Tools:** OpenMC, MOOSE/Cardinal, OpenFOAM, COMSOL.
- **OS and Other Tools:** Linux, Git, LaTeX

STANDARDIZED TEST SCORES

- **IELTS Academic:** Overall Band Score 7.5 [CEFR Level: C1] [TRF]
Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.5.

ACHIEVEMENTS

Bronze Honour, International Astronomy and Astrophysics Competition (IAAC) [PDF] Jul 2022

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- President**, MIST Nuclear Engineering Club (MNEC) Oct 2024 – Jul 2025
- Established cross-year collaboration by launching a structured peer mentorship program: training in core nuclear theory and computational research tools.
 - Organized **Nuclear Tempest '24**, a nuclear science outreach and competition event.
- Club Coordinator**, MIST Nuclear Engineering Club (MNEC) Aug 2023 – Oct 2024
- Led a 14-member volunteer team in executing student-led nuclear science activities and discussions.
 - Coordinated invited talks on academic motivation and undergraduate research experiences.

LANGUAGES

1. **English:** IELTS Academic 7.5 (C1)
2. **Bangla:** Native